

MESSY Hardware-in-the-Loop Extra-Functional Simulator

POLITO | ISOLDE | T4.3 / D4.5 Section 3.1.3 | Version ISOLDE-D4.5-2026.07

Schema Version

0.9

Basic Information

IP Name		MESSY Hardware-in-the-Loop Extra-Functional Simulator
Version		ISOLDE-D4.5-2026.07
Status		Demonstrated
Comment		Draft metadata release derived from ISOLDE D4.5. The value in basicInfo.version identifies this card release, not necessarily the partner's product or repository tag. Partner confirmation is required before publication.
Provider	Item 1 / Organization Name	Politecnico di Torino
Provider	Item 1 / Organization URL	https://www.polito.it/
Provider	Item 1 / Organization Alias	• POLITO
License		Apache-2.0
Repository URL		https://github.com/eml-eda/chessy
Project		ISOLDE
Work Item		T4.3 / D4.5 Section 3.1.3
IP Type	Category	• Software Tool • Domain Specific Simulation Tools • Virtual Prototype • EDA Tools and methods for architecture exploration
IP Type	Domain	• Other
IP Type	Comment	Extra-functional and power-aware system simulation
IP Type	Service	no

System-Level Information

Architecture		RISC-V
Processor		CVA6 and general RISC-V cores
ISAExtension		RV64 for Cheshire workloads
Board		Xilinx ZCU102
SoC		Cheshire host platform
SW Dependencies		SystemC-AMS, Cheshire, RV64 GCC toolchain, Xilinx Vivado, OpenOCD, libjlm and Docker
Requires Specialized Tool		Yes. FPGA and SystemC-AMS co-simulation toolchain.

Interfaces and Artifacts

CLI		• make build-all • make board-flash • make run-all • make help
API		• SystemC-AMS extra-functional model interface • Hardware-in-the-loop communication with the Cheshire FPGA platform
Input Artifacts	Item 1 / Artifact	Cheshire FPGA implementation
Input Artifacts	Item 1 / Format	Bitstream and LTX debug netlist
Input Artifacts	Item 2 / Artifact	RISC-V workload
Input Artifacts	Item 2 / Format	ELF/binary
Input Artifacts	Item 3 / Artifact	Extra-functional models
Input Artifacts	Item 3 / Format	SystemC-AMS
Input Artifacts	Item 4 / Artifact	Optional power models
Input Artifacts	Item 4 / Format	Model parameters
Output Artifacts	Item 1 / Artifact	Extra-functional simulation results
Output Artifacts	Item 1 / Format	Logs and traces
Output Artifacts	Item 2 / Artifact	Hardware-in-the-loop measurements
Output Artifacts	Item 2 / Format	Runtime measurements
Output Artifacts	Item 3 / Artifact	Power/performance observations
Output Artifacts	Item 3 / Format	Analysis data
Deployment		
Bundle		Open-source Git repository with Docker image, Make targets and FPGA integration files
Project-Specific Features		
Purpose		Observe extra-functional properties while software executes on a real RISC-V FPGA platform.
Validated Capabilities		• Cheshire execution on ZCU102 • Host-side SystemC-AMS simulation • Hardware-in-the-loop communication • Reusable make-based setup and test flow
Limitations		• Default deployment targets ZCU102, although the flow can be adapted to other platforms.
Public Repository Status		Public
Source Basis	Deliverable	ISOLDE D4.5 - Software for Hardware Design and Validation Implementation
Source Basis	Section	3.1.3 Integrate system level simulators with extra-functional properties
Source Basis	SW IP Card Model	Schema v0.9 with basicInfo, systemLevelInfo, interfaces, deployment and features
Requirements Summary	Count	6

Requirements Summary	Ids	• COMPUTE-ENGINE-00 • COMPUTE-ENGINE-01 • COMPUTE-ENGINE-02 • COMPUTE-ENGINE-00 • COMPUTE-ENGINE-01 • COMPUTE-ENGINE-02
Metadata Review	Partner Review Required	true
Metadata Review	Items	• Replace the metadata release identifier with the exact product/repository version where available. • Confirm public repository or product URL. • Confirm licensing and redistribution wording. • Add a tool-specific TRL only if the partner has approved it.